

Creation Date 24-Nov-2010

Revision Date 18-Mar-2024

Revision Number 4

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| Product Description: | <b>Bis(2-ethylhexyl) phthalate</b>                          |
| Cat No. :            | <b>A10415</b>                                               |
| Synonyms             | Bis(2-ethylhexyl) phthalate; DOP; Di-2-ethylhexyl phthalate |
| Index No             | 607-317-00-9                                                |
| CAS No               | 117-81-7                                                    |
| EC No                | 204-211-0                                                   |
| Molecular Formula    | C <sub>24</sub> H <sub>38</sub> O <sub>4</sub>              |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Recommended Use      | Laboratory chemicals. Article 56 (3) REACH. Scientific research and development. The substance is used under strictly controlled conditions. |
| Uses advised against | All other uses                                                                                                                               |

### 1.3. Details of the supplier of the safety data sheet

|                |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Company        | Thermo Fisher (Kandel) GmbH<br>Erlenbachweg 2, 76870 Kandel, Germany<br>Tel: +49 (0) 721 84007 280<br>Fax: +49 (0) 721 84007 300<br><br><b>Swiss distributor</b> - Fisher Scientific AG<br>Neuhofstrasse 11, CH 4153 Reinach<br>Tel: +41 (0) 56 618 41 11<br><a href="https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html">https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html</a> |
| E-mail address | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                                                                                                                                                                               |

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

Reproductive Toxicity

Category 1B (H360FD)

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### **Hazard Statements**

H360FD - May damage fertility. May damage the unborn child

### **Precautionary Statements**

P201 - Obtain special instructions before use

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

### **Additional EU labelling**

Restricted to professional users

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Contains a known or suspected endocrine disruptor

Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties

Contains a substance on the National Authorities Endocrine Disruptor Lists

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

### 3.1. Substances

| Component                   | CAS No   | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|-----------------------------|----------|-------------------|----------|---------------------------------------------------|
| Bis(2-ethylhexyl) phthalate | 117-81-7 | EEC No. 204-211-0 | >95      | Repr. 1B (H360FD)                                 |

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Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                      |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                    |
| <b>Skin Contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.                                       |
| <b>Ingestion</b>                          | Call a physician immediately. Clean mouth with water.                                                                                                                                |
| <b>Inhalation</b>                         | Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                     |

### 4.2. Most important symptoms and effects, both acute and delayed

No information available.

### 4.3. Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Water mist may be used to cool closed containers. Chemical foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Vapors may form explosive mixtures with air.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

Vapors are heavier than air and may spread along floors. As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Avoid contact with skin, eyes or clothing. Evacuate personnel to safe areas. Use personal protective equipment as required. Ensure adequate ventilation. Do not touch damaged packages or spilled material.

### 6.2. Environmental precautions

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See Section 12 for additional Ecological Information.

## 6.3. Methods and material for containment and cleaning up

Wear self-contained breathing apparatus and protective suit. Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Do not let this chemical enter the environment.

## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Handle product only in closed system or provide appropriate exhaust ventilation.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1C

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component                   | European Union | The United Kingdom                                                 | France                                     | Belgium                         | Spain                                       |
|-----------------------------|----------------|--------------------------------------------------------------------|--------------------------------------------|---------------------------------|---------------------------------------------|
| Bis(2-ethylhexyl) phthalate |                | STEL: 10 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr | TWA / VME: 5 mg/m <sup>3</sup> (8 heures). | TWA: 5 mg/m <sup>3</sup> 8 uren | TWA / VLA-ED: 5 mg/m <sup>3</sup> (8 horas) |

| Component                   | Italy | Germany                                                                                                                                           | Portugal                         | The Netherlands | Finland                                                                          |
|-----------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------|----------------------------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate |       | TWA: 2 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 4 mg/m <sup>3</sup> Haut | TWA: 5 mg/m <sup>3</sup> 8 horas |                 | TWA: 5 mg/m <sup>3</sup> 8 tunteina<br>STEL: 10 mg/m <sup>3</sup> 15 minuutteina |

| Component | Austria | Denmark | Switzerland | Poland | Norway |
|-----------|---------|---------|-------------|--------|--------|
|-----------|---------|---------|-------------|--------|--------|

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|                             |                                                                                                                                 |                                                                           |                                                 |                                                                               |                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate | MAK-KZGW: 50 mg/m <sup>3</sup> 15 Minuten<br>MAK-KZGW: 5 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 5 mg/m <sup>3</sup> 8 Stunden | TWA: 3 mg/m <sup>3</sup> 8 timer<br>STEL: 6 mg/m <sup>3</sup> 15 minutter | Haut/Peau<br>TWA: 2 mg/m <sup>3</sup> 8 Stunden | STEL: 5 mg/m <sup>3</sup> 15 minutach<br>TWA: 1 mg/m <sup>3</sup> 8 godzinach | TWA: 1 mg/m <sup>3</sup> 8 timer<br>STEL: 3 mg/m <sup>3</sup> 15 minutter. value calculated |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

| Component                   | Bulgaria                                                   | Croatia                                                                                | Ireland | Cyprus | Czech Republic                                                                               |
|-----------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------|---------|--------|----------------------------------------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate | TWA: 5.0 mg/m <sup>3</sup><br>STEL : 5.0 mg/m <sup>3</sup> | TWA-GVI: 5 mg/m <sup>3</sup> 8 satima.<br>STEL-KGVI: 10 mg/m <sup>3</sup> 15 minutama. |         |        | TWA: 5 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 10 mg/m <sup>3</sup> toxic for reproduction |

| Component                   | Estonia                                                                                                                                   | Gibraltar | Greece                                                 | Hungary                                                                                                                        | Iceland                                                                                                                                                                                                        |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate | TWA: 3 mg/m <sup>3</sup> 8 tundides. regulated under CAS 117-84-0<br>STEL: 5 mg/m <sup>3</sup> 15 minutites. regulated under CAS 117-84-0 |           | STEL: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> | STEL: 4 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 2 mg/m <sup>3</sup> 8 órában. AK<br>lehetséges borón keresztüli felszívódás | TWA: 3 mg/m <sup>3</sup> 8 klukkustundum. same limit value expressed in mg/m <sup>3</sup> shall also be applied for those phthalates for which no limit value has been defined<br>Ceiling: 6 mg/m <sup>3</sup> |

| Component                   | Latvia | Lithuania                                                  | Luxembourg | Malta | Romania |
|-----------------------------|--------|------------------------------------------------------------|------------|-------|---------|
| Bis(2-ethylhexyl) phthalate |        | TWA: 3 mg/m <sup>3</sup> IPRD<br>STEL: 5 mg/m <sup>3</sup> |            |       |         |

| Component                   | Russia                   | Slovak Republic                                          | Slovenia                                                                                                              | Sweden                                                                                    | Turkey |
|-----------------------------|--------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|
| Bis(2-ethylhexyl) phthalate | MAC: 1 mg/m <sup>3</sup> | Ceiling: 5 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> 8 urah inhalable fraction<br>Koža<br>STEL: 4 mg/m <sup>3</sup> 15 minutah inhalable fraction | Indicative STEL: 5 mg/m <sup>3</sup> 15 minuter<br>TLV: 3 mg/m <sup>3</sup> 8 timmar. NGV |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                    | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|----------------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Bis(2-ethylhexyl) phthalate 117-81-7 ( >95 ) |                              |                                 |                                | DNEL = 3.4mg/kg bw/day            |

| Component                                    | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|----------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Bis(2-ethylhexyl) phthalate 117-81-7 ( >95 ) |                                  |                                     |                                    | DNEL = 1.6mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

See values below.

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| Component                                       | Fresh water | Fresh water sediment           | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)        |
|-------------------------------------------------|-------------|--------------------------------|--------------------|------------------------------------|---------------------------|
| Bis(2-ethylhexyl) phthalate<br>117-81-7 ( >95 ) |             | PNEC = 100mg/kg<br>sediment dw |                    | PNEC = 201mg/L                     | PNEC = 13mg/kg<br>soil dw |

| Component                                       | Marine water | Marine water sediment         | Marine water Intermittent | Food chain              | Air |
|-------------------------------------------------|--------------|-------------------------------|---------------------------|-------------------------|-----|
| Bis(2-ethylhexyl) phthalate<br>117-81-7 ( >95 ) |              | PNEC = 20mg/kg<br>sediment dw |                           | PNEC = 3.3mg/kg<br>food |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material                                      | Breakthrough time                    | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|--------------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers<br>recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

#### Physical State

Liquid

#### Appearance

Light yellow

#### Odor

Odorless

#### Odor Threshold

No data available

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Melting Point/Range                     | -50 °C / -58 °F          |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | 384 °C / 723.2 °F        | @ 760 mmHg                        |
| Flammability (liquid)                   | No data available        |                                   |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | Lower 0.1                |                                   |
| Flash Point                             | 195 °C / 383 °F          | Method - No information available |
| Autoignition Temperature                | 390 °C / 734 °F          |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No information available |                                   |
| Viscosity                               | 80 mPa.s at 20 °C        |                                   |
| Water Solubility                        | Virtually insoluble      |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Bis(2-ethylhexyl) phthalate             | 5.03                     |                                   |
| Vapor Pressure                          | 1.8 mbar @ 200 °C        |                                   |
| Density / Specific Gravity              | 0.981                    |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Vapor Density                           | 13.46 (Air = 1.0)        | (Air = 1.0)                       |
| Particle characteristics                | Not applicable (liquid)  |                                   |

## 9.2. Other information

|                   |            |
|-------------------|------------|
| Molecular Formula | C24 H38 O4 |
| Molecular Weight  | 390.55     |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | No information available.                |

### 10.4. Conditions to avoid

Incompatible products.

### 10.5. Incompatible materials

Strong oxidizing agents.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

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## Inhalation

Based on available data, the classification criteria are not met

| Component                   | LD50 Oral              | LD50 Dermal               | LC50 Inhalation                            |
|-----------------------------|------------------------|---------------------------|--------------------------------------------|
| Bis(2-ethylhexyl) phthalate | LD50 = 30 g/kg ( Rat ) | LD50 = 25 g/kg ( Rabbit ) | LC50 > 10620 mg/m <sup>3</sup> ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component                   | EU | UK | Germany | IARC     |
|-----------------------------|----|----|---------|----------|
| Bis(2-ethylhexyl) phthalate |    |    |         | Group 2B |

(g) reproductive toxicity;  
Reproductive Effects Category 1B  
Teratogenicity Product is or contains a chemical which is a known or suspected reproductive hazard. Teratogenic effects have occurred in experimental animals.

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Other Adverse Effects See actual entry in RTECS for complete information

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

Endocrine Disrupting Properties  
Assess endocrine disrupting properties for human health  
Substance identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 Contains a substance on the National Authorities Endocrine Disruptor Lists

| Component                                       | EU National Authorities Endocrine Disruptor Lists - Health |
|-------------------------------------------------|------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate<br>117-81-7 ( >95 ) | List I                                                     |

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity



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**Ecotoxicity effects** Do not empty into drains.

| Component                   | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Water Flea                                                                      | Freshwater Algae                                                                                                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate | LC50: > 0.32 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: > 0.32 mg/L, 96h semi-static (Oryzias latipes)<br>LC50: > 0.32 mg/L, 96h semi-static (Brachydanio rerio)<br>LC50: > 0.32 mg/L, 96h semi-static (Poecilia reticulata)<br>LC50: > 0.67 mg/L, 96h flow-through (Oryzias latipes)<br>LC50: > 100 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 0.27 - 0.67 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 0.16 mg/L, 96h static (Pimephales promelas)<br>LC50: > 0.200 mg/L, 96h static (Lepomis macrochirus)<br>LC50: > 0.200 mg/L, 96h flow-through (Lepomis macrochirus) | LC50: = 9.4 mg/L, 48h (Daphnia magna)<br>EC50: > 0.16 mg/L, 48h (Daphnia magna) | EC50: > 0.1 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: > 0.1 mg/L, 96h (Pseudokirchneriella subcapitata)<br>EC50: > 130 mg/L, 72h (Desmodesmus subspicatus) |

| Component                   | Microtox                                                                  | M-Factor |
|-----------------------------|---------------------------------------------------------------------------|----------|
| Bis(2-ethylhexyl) phthalate | EC50 = 800 mg/L 15 min<br>EC50 = 800 mg/L 30 min<br>EC50 = 800 mg/L 5 min |          |

**12.2. Persistence and degradability** Expected to be biodegradable  
**Persistence** May persist.

**12.3. Bioaccumulative potential** Product has a high potential to bioconcentrate

| Component                   | log Pow | Bioconcentration factor (BCF) |
|-----------------------------|---------|-------------------------------|
| Bis(2-ethylhexyl) phthalate | 5.03    | 1 - 29.7 dimensionless        |

**12.4. Mobility in soil** The product is insoluble and floats on water . Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

**12.5. Results of PBT and vPvB assessment** Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

## 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information**  
**Assess endocrine disrupting properties for the environment**

Substance identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605. Contains a substance on the National Authorities Endocrine Disruptor Lists.

| Component                   | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|-----------------------------|------------------------------------------|--------------------------------------------------|
| Bis(2-ethylhexyl) phthalate | Group I Chemical                         | High Exposure Concern                            |

| Component                                       | EU National Authorities Endocrine Disruptor Lists - Environment | Japan - Endocrine Disruptor Information |
|-------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------|
| Bis(2-ethylhexyl) phthalate<br>117-81-7 ( >95 ) | List I                                                          |                                         |

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**12.7. Other adverse effects**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**

Not regulated

**14.1. UN number**

**14.2. UN proper shipping name**

**14.3. Transport hazard class(es)**

**14.4. Packing group**

**ADR**

Not regulated

**14.1. UN number**

**14.2. UN proper shipping name**

**14.3. Transport hazard class(es)**

**14.4. Packing group**

**IATA**

Not regulated

**14.1. UN number**

**14.2. UN proper shipping name**

**14.3. Transport hazard class(es)**

**14.4. Packing group**

**14.5. Environmental hazards**

No hazards identified

**14.6. Special precautions for user**

No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments**

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories**

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Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                   | CAS No   | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------------------|----------|-----------|--------|-----|-------|------|----------|------|------|
| Bis(2-ethylhexyl) phthalate | 117-81-7 | 204-211-0 | -      | -   | X     | X    | KE-02196 | X    | X    |

| Component                   | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------------------|----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Bis(2-ethylhexyl) phthalate | 117-81-7 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component                   | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization                                                                                                                                                                                                                     | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                     | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)                                                                                          |
|-----------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate | 117-81-7 | Toxic for reproduction<br>Category 1B Article 57<br>Application date: August 21, 2013<br>Sunset date: February 21, 2015<br>Exemption - Uses in the immediate packaging of medicinal products covered under Regulation (EC) No 726/2004 Directive 2001/82/EC and/or Directive 2011/83/EC | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 51[a].<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 204-211-0 - Toxic for reproduction, Article 57c; Endocrine disrupting properties, Article 57f - environment; Endocrine disrupting properties, Article 57f - human health |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

### REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Seveso III Directive (2012/18/EC)

| Component                   | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate | 117-81-7 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

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**WGK Classification** See table for values

| Component                   | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------------------|---------------------------------------|-------------------------|
| Bis(2-ethylhexyl) phthalate | WGK1                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                                       | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Bis(2-ethylhexyl) phthalate<br>117-81-7 ( >95 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H360FD - May damage fertility. May damage the unborn child

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

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|                  |                                                    |
|------------------|----------------------------------------------------|
| Prepared By      | Health, Safety and Environmental Department        |
| Creation Date    | 24-Nov-2010                                        |
| Revision Date    | 18-Mar-2024                                        |
| Revision Summary | New emergency telephone response service provider. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**